

Electrophilic addition to alkenes

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Connections

➡ Building on

- Elimination reactions that form alkenes [ch17](#)
- Stability of carbocations, and their reactions during the S_N1 reaction [ch15](#)

Arriving at

- Reactions of simple, unconjugated alkenes with *electrophiles*
- Converting $C=C$ double bonds to other functional groups by electrophilic addition
- How to predict which end of an unsymmetrical alkene reacts with the electrophile
- Stereoselective, stereospecific, and regioselective reactions of alkenes
- How to make alkyl halides, epoxides, alcohols, and ethers through electrophilic addition
- How to cleave an alkene into two carbonyl compounds

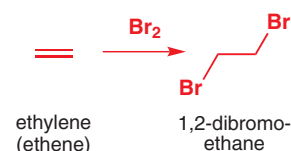
➡ Looking forward to

- Electrophilic addition to alkenes carrying oxygen substituents (enols and enolates) [ch20](#)
- Electrophilic addition to aromatic rings [ch21](#)
- Nucleophilic additions to electron-deficient alkenes [ch22](#)
- Reactions of alkenes by pericyclic reactions [ch34](#)
- Rearrangement reactions [ch36](#)

Alkenes react with bromine

Bromine (Br_2) is brown, and one of the classic tests for alkenes is that they turn a brown aqueous solution of bromine colourless. Alkenes decolourize bromine water: alkenes react with bromine. The product of the reaction is a dibromoalkane, and the reaction on the right shows what happens with the simplest alkene, ethylene (ethene).

In order to understand this reaction, and the other similar ones you will meet in this chapter, you need to think back to Chapter 5, where we started talking about reactivity in terms of nucleophiles and electrophiles. As soon as you see a new reaction, you should immediately think to yourself, ‘Which reagent is the nucleophile; which reagent is the electrophile?’ Evidently, neither the alkene nor bromine is charged, but Br_2 has a low-energy empty orbital (the $Br-Br$ σ^*), and is therefore an electrophile. The $Br-Br$ bond is exceptionally weak, and bromine reacts with many nucleophiles like this.



In the reaction with ethylene, the alkene must be the nucleophile, and its HOMO is the $C=C$ π bond. Other simple alkenes are similarly electron-rich and they typically act as *nucleophiles* and attack *electrophiles*.

- Simple, unconjugated alkenes are nucleophilic and react with electrophiles.

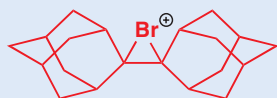
Online support. The icon  in the margin indicates that accompanying interactive resources are provided online to help your understanding: just type www.chemtube3d.com/clayden/123 into your browser, replacing **123** with the number of the page where you see the icon. For pages linking to more than one resource, type **123-1**, **123-2** etc. (replacing **123** with the page number) for access to successive links.

When it reacts with Br_2 , the alkene's filled π orbital (the HOMO) will interact with the bromine's empty σ^* orbital to give a product. But what will that product be? Look at the orbitals involved.

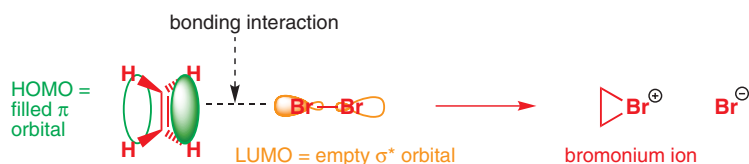


How do we know bromonium ions exist?

Very hindered alkenes form bromonium ions that are resistant to nucleophilic attack. In this very hindered case, the bromonium ion is sufficiently stable to be characterized by X-ray crystallography.



The highest electron density in the π orbital is right in the middle, between the two carbon atoms, so this is where we expect the bromine to attack. The only way the π HOMO can interact in a bonding manner with the σ^* LUMO is if the Br_2 approaches end-on—and this is how the product forms. The symmetrical three-membered ring product is called a bromonium ion.



How shall we draw curly arrows for the formation of the bromonium ion? We have a choice. The simplest way is just to show the middle of the π bond attacking $\text{Br}-\text{Br}$, mirroring what we know happens with the orbitals.

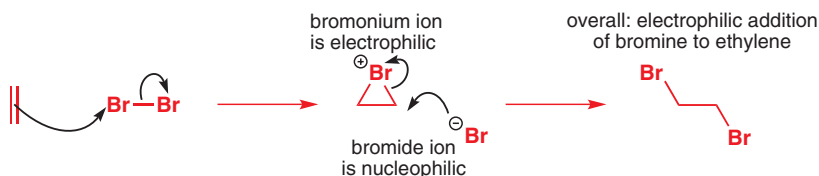


But there is a problem with this representation: because only *one* pair of electrons is moving, we can't form *two* new $\text{C}-\text{Br}$ bonds. We should really then represent the $\text{C}-\text{Br}$ bonds as partial bonds. Yet the bromonium ion is a real intermediate with two proper $\text{C}-\text{Br}$ bonds (the box in the margin presents evidence of this). So an alternative way of drawing the arrows is to involve a lone pair on bromine.



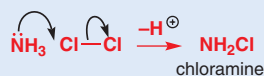
We think the first way represents more accurately the key orbital interaction involved, and we shall use that one, but the second is acceptable too.

Of course, the *final* product of the reaction isn't the bromonium ion. The second step of the reaction follows on at once: the bromonium ion is itself an electrophile, and it reacts with the bromide ion lost from the bromine in the addition step. We can now draw the correct mechanism for the whole reaction, which is termed electrophilic addition to the double bond, because bromine (Br_2) is an electrophile. Overall, the molecule of bromine *adds across* the double bond of the alkene.



Chloramines

Have you ever wondered why conventional wisdom (and manufacturers' labels) warns against mixing different types of cleaning agent? The danger arises from nucleophilic attack on another electrophilic halogen, chlorine. Some cleaning solutions contain chlorine (bleach, to kill moulds and bacteria, usually for the bathroom) while others contain ammonia (to dissolve fatty deposits, usually for the kitchen). Ammonia is nucleophilic, chlorine electrophilic, and the products of their reaction are the highly toxic and explosive chloramines NH_2Cl , NHCl_2 , and NCl_3 .

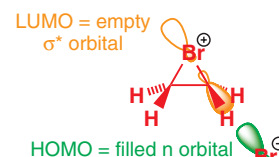


Interactive mechanism for reaction of ethylene with bromine

➡ Compare the second step with the way nucleophiles attack epoxides, Chapter 15, p. 354.

Attack of Br on a bromonium ion is a normal S_N2 substitution—the key orbitals involved are the HOMO of the bromide and the σ^* of one of the two carbon–bromine bonds in the strained three-membered ring. As with all S_N2 reactions, the nucleophile maintains maximal overlap with the σ^* by approaching in line with the leaving group but from the opposite side, resulting in inversion at the carbon that is attacked. The stereochemical outcome of more complicated reactions (discussed below) is important evidence for this overall reaction mechanism.

You may wonder why the bromine attacks a carbon atom in the bromonium ion rather than the positively charged bromine atom. Well, in fact, it can do this as well, but the result is just regeneration of bromine and the alkene: the first step of the reaction is reversible.

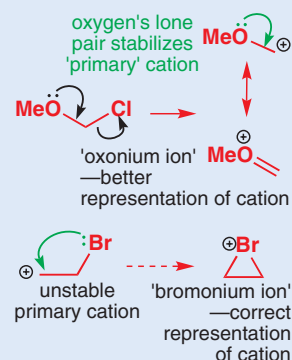


Another way of thinking about bromonium ions

You can think of the bromonium ion as a carbocation that has been stabilized by interaction with a nearby bromine atom. You have seen a similar effect with oxygen—this 'oxonium ion' was an intermediate, for example, in the S_N1 substitution of MOM chloride on p. 338 of Chapter 15.

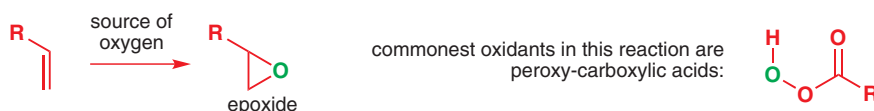
The bromine is one atom further away but, with bromine being lower in the periodic table and having more diffuse lone pairs, it can have a similar stabilizing effect, despite the angle strain in a three-membered ring.

The two types of stabilization are not equivalent: the cation and the bromonium ion are different molecules with different shapes, while the two representations of the oxonium ion are just that—they aren't different molecules. This stabilization of an adjacent cationic centre by a heteroatom with at least one lone pair to form a three-membered ring intermediate is not restricted to bromine or the other halogens, but is also an important aspect of the chemistry of compounds containing oxygen, sulfur, or selenium, as you will see in Chapter 27.

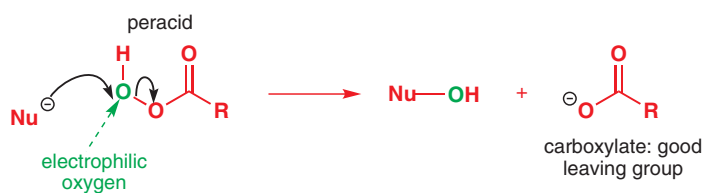


Oxidation of alkenes to form epoxides

The electrophilic addition of bromine to alkenes is an oxidation. The starting alkene is equivalent in oxidation level to an alcohol, but the product has *two* carbons at the alcohol oxidation level—the elimination reactions of dibromides to give alkynes that you met in Chapter 17 (p. 398) should convince you of this. There are a number of other oxidants containing electrophilic oxygen atoms that react with nucleophilic alkenes to produce epoxides (oxiranes). You can view epoxides as the oxygen analogues of bromonium ions, but unlike most bromonium ions they are quite stable.



The simplest epoxide, ethylene oxide (or oxirane itself), can be produced on the tonne scale by the direct oxidation of ethene with oxygen at high temperature over a silver catalyst. These conditions are hardly suitable for general laboratory use, and the most commonly used epoxidizing agents are peroxy-carboxylic acids. These peroxy-acids (or peracids) have an extra oxygen atom between the carbonyl group and their acidic hydrogen—they are esters of hydrogen peroxide (H_2O_2). They are rather less acidic than carboxylic acids because their conjugate base is no longer stabilized by delocalization into the carbonyl group. But they are electrophilic at the oxygen shown here in green because attack there by a nucleophile displaces carboxylate, a good leaving group. The LUMO of a peroxy-carboxylic acid is the σ^* orbital of the weak O–O bond.

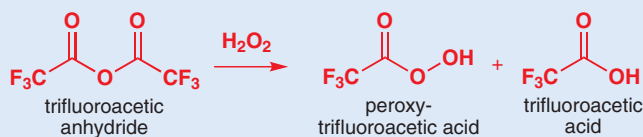


■ You have met epoxides being formed by intramolecular substitution reactions, but the oxidation of alkenes is a much more important way of making them. Their alternative name derives from a systematic way of naming rings: 'ox' for the O atom, 'ir' for the three-membered ring, and 'ane' for full saturation. You may meet oxetane (remember the oxaphosphetane in the Wittig reaction, Chapter 11, p. 238) and, while THF is never called oxolane, dioxolane is another name for five-membered cyclic acetals.

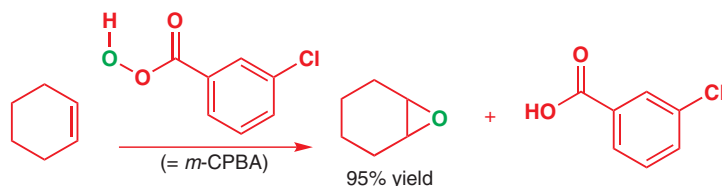


Making peroxy-acids

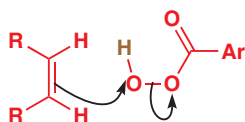
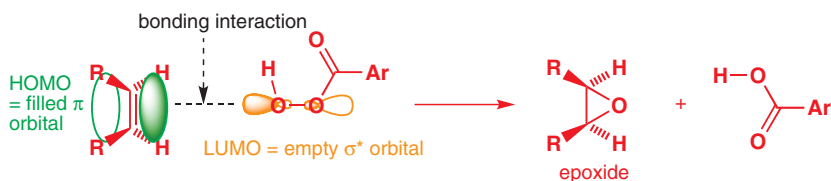
Peroxy-acids are prepared from the corresponding acid anhydride and high-concentration hydrogen peroxide. In general, the stronger the parent acid, the more powerful the oxidant (because the carboxylate is a better leaving group): one of the most powerfully oxidizing peroxy-acids is peroxy-trifluoroacetic acid. Hydrogen peroxide, at very high concentrations (> 80%), is potentially explosive and difficult to transport.



The most commonly used peroxy-acid is known as *m*-CPBA, or *meta*-chloroperoxybenzoic acid. *m*-CPBA is a safely crystalline solid. Here it is, reacting with cyclohexene, to give the epoxide in 95% yield.

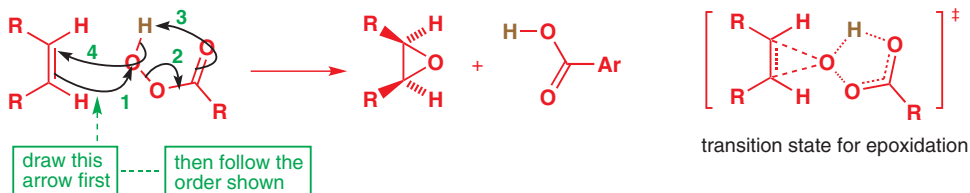


As you would expect, the nucleophilic alkene attacks the peroxy-acid from the centre of the HOMO, its π orbital. First, here is the orbital involved.



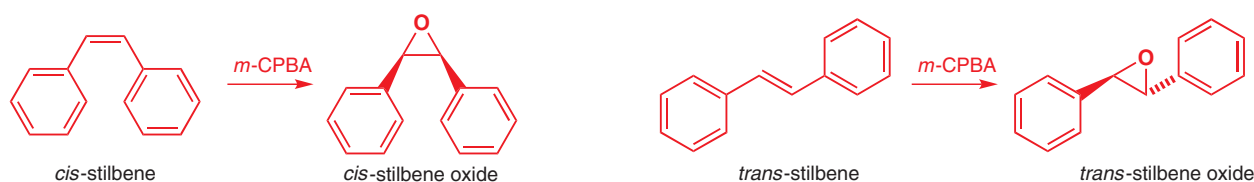
And now the curly arrow mechanism. The essence of the mechanism is attack by the π orbital of the alkene on the weak, polarized, electrophilic O—O bond, which we can represent most simply as shown in the margin. But, in the real reaction, a proton (shown in brown in this mechanism) has transferred from the epoxide oxygen to the carboxylic acid by-product. You can represent this all in one step if you draw the arrows carefully. Start with the nucleophilic π bond: send the electrons on to oxygen, breaking O—O and forming a new carbonyl bond. Use those electrons to pick up the proton, and use the old O—H bond's electrons to make the second new C—O bond. Don't be put off by the spaghetti effect—each arrow is quite logical when you think the mechanism through. The transition state for the reaction makes the bond-forming and -breaking processes clearer.

Interactive mechanism for epoxidation of ethylene



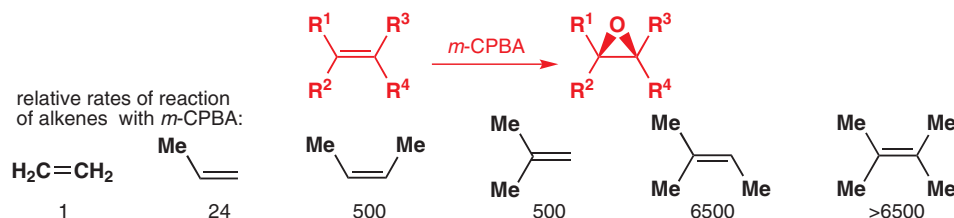
Epoxidation is stereospecific

Because both new C—O bonds are formed on the same face of the alkene's π bond, the geometry of the alkene is reflected in the stereochemistry of the epoxide. The reaction is therefore stereospecific. Here are two examples demonstrating this: *cis*-alkene gives *cis*-epoxide and *trans*-alkene gives *trans*-epoxide.

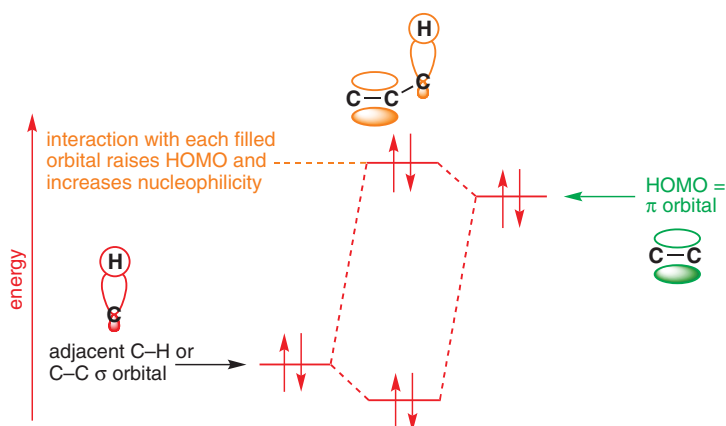


More substituted alkenes epoxidize faster

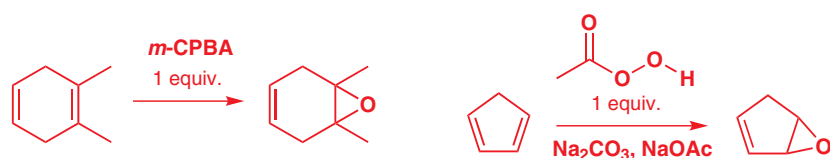
Peracids give epoxides from alkenes with any substitution pattern (except ones conjugated with electron-withdrawing groups, for which a different reagent is required: see Chapter 22), but the chart below shows how the rate varies according to the number of substituents on the double bond.



Not only are more substituted double bonds more stable (as you saw in Chapter 17), but they are more nucleophilic. We showed you in Chapter 15 that alkyl groups are electron-donating because they stabilize carbocations. This same electron-donating effect raises the energy of the HOMO of a double bond and makes it more nucleophilic. You can think of it this way: every C–C or C–H bond that can allow its σ orbital to interact with the π orbital of the alkene will raise the HOMO of the alkene slightly, as shown by the energy level diagram. The more substituents the alkene has, the more the energy is raised.



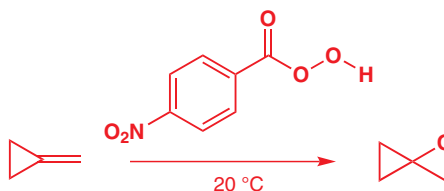
The differences in reactivity between alkenes of different substitution patterns can be exploited to produce the epoxide only of the more reactive alkene of a pair, provided the supply of oxidant is limited. In the first example below, a tetrasubstituted alkene reacts in preference to a *cis* disubstituted one. Even when two alkenes are equally substituted, the effect of epoxidizing one of them is to reduce the nucleophilicity of the second (the new oxygen atom is electron-withdrawing, and dienes are in general more nucleophilic than alkenes: see below). The monoepoxide of cyclopentadiene is a useful intermediate and can be prepared by direct epoxidation of the diene under buffered conditions.



■ The sodium carbonate/sodium acetate added here is a *buffer*, used to prevent the reaction mixture becoming too acidic—remember, the carboxylic acid is a by-product of the epoxidation. Some epoxides are unstable in acid, as we shall see shortly.

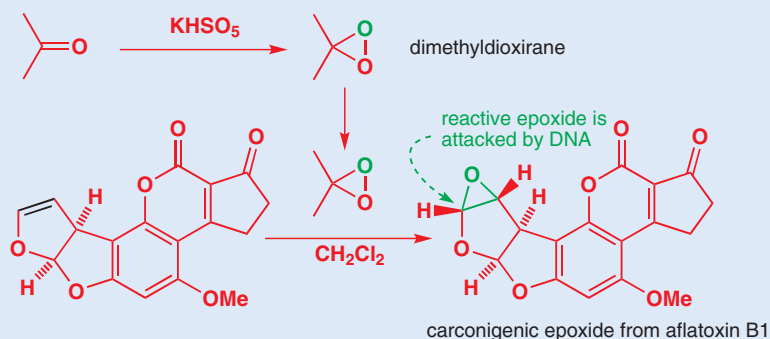
■ *Spiro* compounds have two rings joined at a single atom. Compare **fused** rings (joined at two adjacent atoms) and **bridged** rings (joined at two non-adjacent atoms) (see p. 653).

p-Nitroperoxybenzoic acid is dangerously explosive, and it is sufficiently reactive to produce this remarkable and highly strained *spiro* epoxide (oxaspiropentane), which was made in order to study its reactions with nucleophiles.

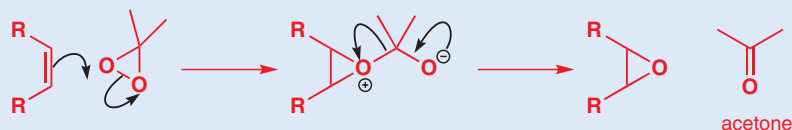


Dimethyldioxirane and carcinogenic epoxides

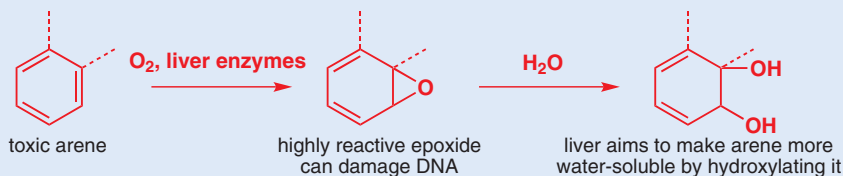
Certain fungi, especially the mould *Aspergillus* sp. (which grows on damp grain), produce a group of the most carcinogenic substances known to man, the aflatoxins. One of the toxins (which are, of course, entirely natural) is metabolized in the human body to the epoxide shown below. Some chemists in the USA decided to synthesize this epoxide to investigate its reaction with DNA, hoping to discover exactly how it causes cancer. The epoxide is far too reactive to be made using a peroxy-acid (because of the acidic by-product), and instead these chemists used a reagent called dimethyldioxirane.



Dimethyldioxirane is made by oxidizing acetone with KHSO_5 , but is too reactive to be stored for more than a short period in solution. After it has transferred an oxygen atom in the epoxidation step, only innocuous acetone is left, as shown by the mechanism below.

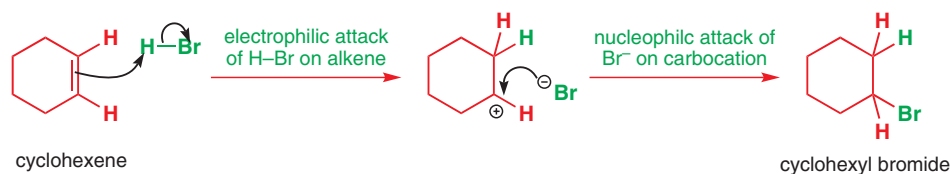


The liver is home to a wide variety of enzymes that carry out oxidation—the aim is to make unwanted water-insoluble molecules more polar and therefore soluble by peppering them with hydroxyl groups. Unfortunately, some of the intermediates in the oxidation processes are highly reactive epoxides that damage DNA. This is the means by which aromatic hydrocarbons may cause cancer, for example. Note that it is very hard to oxidize benzene by chemical (rather than biological) methods.



Electrophilic addition to unsymmetrical alkenes is regioselective

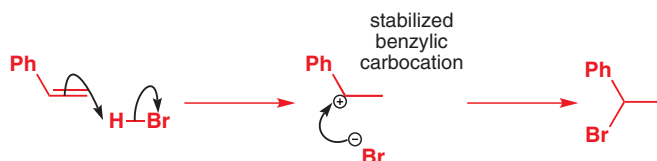
In epoxidation reactions, and in electrophilic additions of bromine, each end of the alkene is joined to the same sort of atom (Br or O). But in the addition reactions of other electrophiles, H–Br for example, there is a choice: which carbon gets the H and which gets the Br? You will need to be able to predict, and to explain, reactions of unsymmetrical alkenes with HBr, but we should start by looking at the reaction with a symmetrical alkene—cyclohexene. This is what happens. When H–Br reacts as an electrophile, it is attacked at H, losing Br[−]. Unlike a bromine atom, a hydrogen atom can't form a three-membered ring cation—it has no lone pairs to use. So electrophilic addition of a proton (which is what this is) to an alkene gives a product best represented as a carbocation. This carbocation rapidly reacts with the bromide ion just formed. Overall, H–Br adds across the alkene. This is a useful way of making simple alkyl bromides.



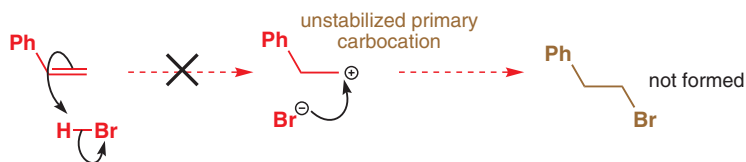
Here are two more syntheses of alkyl bromides, but this time we need to ask our question about which end of the alkene is attacked because the alkenes are unsymmetrical (they have different substituents at each end). First, the results.



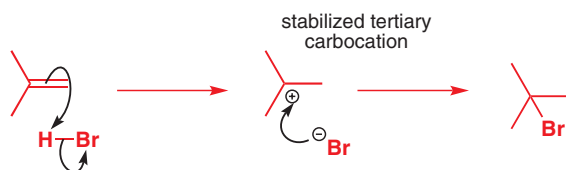
In each case, the bromine atom ends up on the more substituted carbon, and the mechanism explains why. There are two possible outcomes for protonation of styrene by HBr, but you should immediately be able to spot which is preferred, even if you don't know the outcome of the reaction. Protonation at one end gives a stabilized benzylic cation, with its positive charge delocalized into the benzene ring.



Protonation at the other end would give a highly unstable primary cation, and therefore does not take place.



You get the same result with isobutene (2-methylpropene): the more stable tertiary cation leads to the product; the alternative primary cation is not formed.

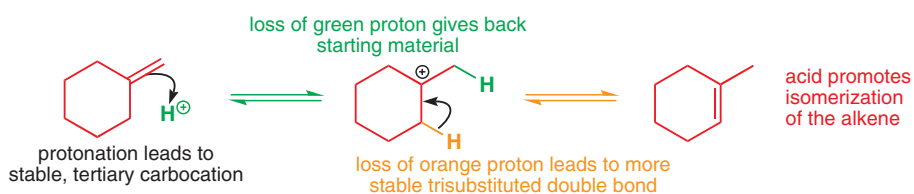


Markovnikov's rule

There is a traditional guideline called Markovnikov's rule for electrophilic additions of H–X to alkenes, which can be stated as: 'The hydrogen ends up attached to the carbon of the double bond that had more hydrogens to start with.' We don't suggest you learn this rule, although you may hear it referred to. As with all 'rules' it is much more important to understand the reason behind it. For example, *you* can now predict the product of the reaction below. With all due respect, Markovnikov couldn't.

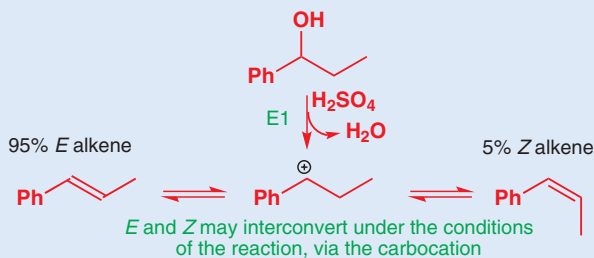


The protonation of alkenes to give carbocations is quite general. The carbocations may trap a nucleophile, as you have just seen, or they may simply lose a proton to give back an alkene. This is just the same as saying the protonation is reversible, but it needn't be the same proton that is lost. A more stable alkene may be formed by losing a different proton, which means that acid can catalyse the isomerization of alkenes—both between *Z* and *E* geometrical isomers and between regioisomers.

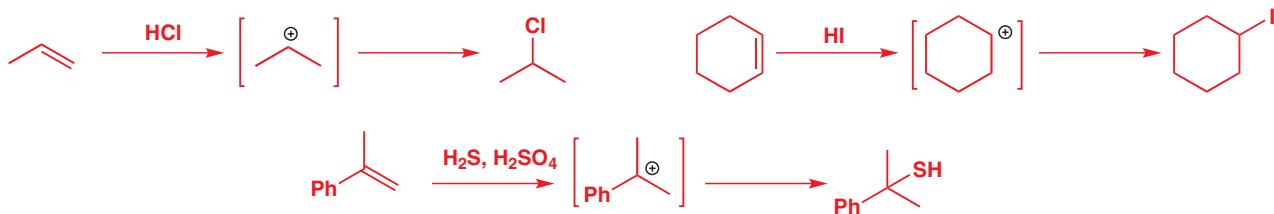


E1 and isomerization

The isomerization of alkenes in acid is probably a good part of the reason why E1 eliminations in acid generally give *E* alkenes. In Chapter 17 we explained how *kinetic* control could lead to *E* alkenes: interconversion of *E* and *Z* alkenes under the conditions of the reaction allows the *thermodynamic* product to prevail. This was also discussed in Chapter 12.

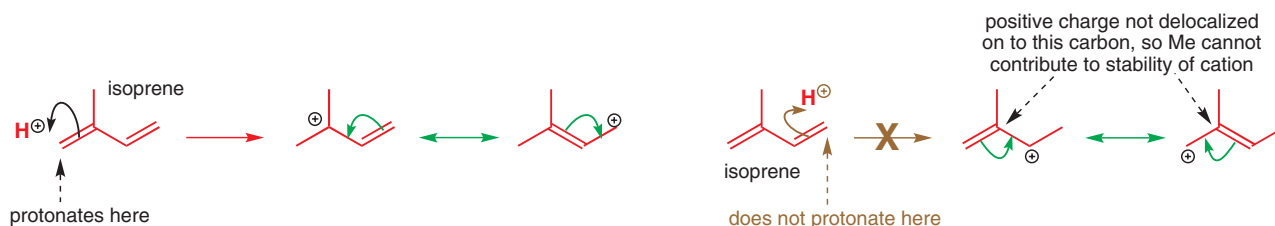


Other nucleophiles may also intercept the cation, for example alkenes can be treated with HCl to form alkyl chlorides, with HI to form alkyl iodides, and with H₂S to form thiols.



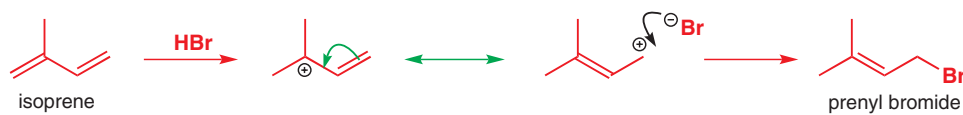
Electrophilic addition to dienes

Earlier in the chapter you saw the epoxidation of a diene to give a monoepoxide: only one of the double bonds reacted. This is quite a usual observation: dienes are more nucleophilic than isolated alkenes. This is easy to explain by looking at the relative energy of the HOMO of an alkene and a diene—this discussion is on p. 138 of Chapter 7. Dienes are therefore very susceptible to protonation by acid to give a cation. This is what happens when 2-methylbuta-1,3-diene (isoprene) is treated with acid. Protonation gives a stable delocalized allylic cation.

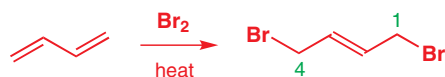


Why protonate this double bond and not the other one? The cation you get by protonating the other double bond is also allylic, but it cannot benefit from the additional stabilization from the methyl group because the positive charge is not delocalized on to the carbon carrying the methyl.

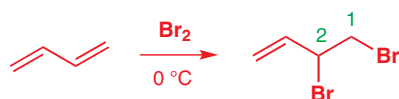
If the acid is HBr, then nucleophilic attack by Br on the cation follows. The cation is attacked at the less hindered end to give the important compound prenyl bromide. This is very much the sort of reaction you met in Chapter 15—it is the second half of an S_N1 substitution reaction on an allylic compound.



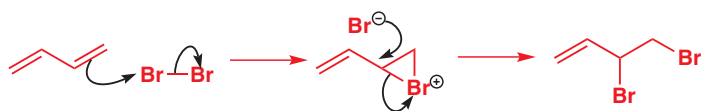
Overall, the atoms H and Br are added to the ends of the diene system. The same appears to be the case when dienes are brominated with Br_2 .



Changing the conditions slightly gives a different outcome. If the reaction is done at lower temperatures, the bromine just adds across one of the double bonds to give a 1,2-dibromide.



This compound turns out to be the kinetic product of the bromination reaction. The 1,4-dibromide is formed only when the reaction is heated, and is the thermodynamic product. The mechanism is electrophilic attack on the diene to give a bromonium ion, which bromide opens to give the dibromide. We have shown the bromide attacking the more substituted end of the bromonium ion—although we can't know this for sure (attack at either end gives the same product), you are about to see (in the next section) evidence that this is the usual course of reactions of unsymmetrical bromonium ions.



➡ If you need reminding about kinetic and thermodynamic control, look back at p. 264, Chapter 12.

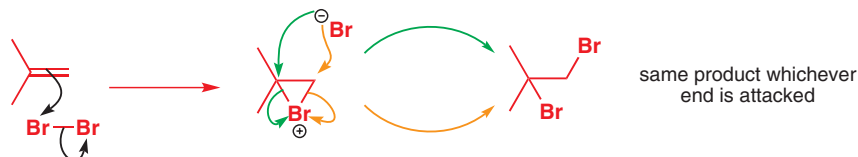
This 1,2-dibromide can still react further because it can undergo nucleophilic substitution. Bromide is a good nucleophile and a good leaving group and, with an allylic system like this, an S_N1 reaction can take place in which both the nucleophile and the leaving group are bromide. The intermediate is a cation, but here the carbocation is disguised as the bromonium ion because bromine's lone pair can help stabilize the positive charge. Bromide can attack where it left, returning to starting material, but it can also attack the far end of the allylic system, giving the 1,4-dibromide. The steps are all reversible at higher temperatures, so the fact that the 1,4-dibromide is formed under these conditions must mean it is more stable than the 1,2-dibromide. It is not hard to see why: it has a more substituted double bond and the two large bromine atoms are further apart.



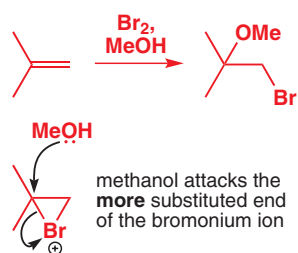
Interactive mechanism for bromination of butadiene

Unsymmetrical bromonium ions open regioselectively

We ignored the issue of symmetry in the alkene when we discussed the bromination of alkenes because even unsymmetrical alkenes give the same 1,2-dibromides, whichever way the bromide attacks the bromonium ion.

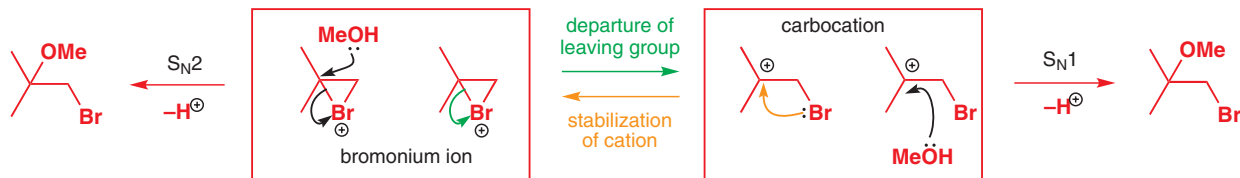


But when a bromination is done in a nucleophilic solvent—water or methanol, for example—solvent molecules compete with the bromide to open the bromonium ion. As you know, alcohols are much worse nucleophiles than bromide but, because the concentration of solvent is so high (remember—the concentration of water in water is 55 M), the solvent gets there first most of the time. This is what happens when isobutene is treated with bromine in methanol. An ether is formed by attack of methanol only at the *more substituted* end of the bromonium ion. When a functional group can react in more than one position, the choice is known as the *regioselectivity* of the reaction. We will return to the concept of regioselectivity in Chapter 24.

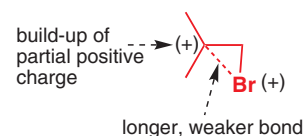


Methanol is attacking the bromonium ion where it is most hindered, so there must be some effect at work more powerful than steric hindrance. One way of looking at this is to reconsider our assumption that bromonium ion opening is an S_N2 process. Here, it hardly looks S_N2 . We have a tertiary centre, so naturally you expect S_N1 , via the cation below. But we have already said that cations like this can be stabilized by formation of the three-membered bromonium ion and, if we let this happen, we have to attack the bromonium ion, which gets us back to where we started: an S_N2 mechanism!

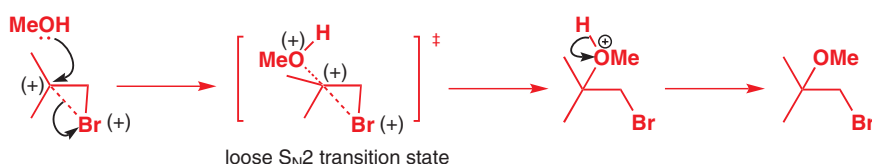
two limiting mechanisms for substitution on bromonium ion



The answer to the conundrum is that substitution reactions don't always go by pure S_N1 or pure S_N2 mechanisms: sometimes the mechanism is somewhere in between. Perhaps the leaving group starts to leave, creating a partial positive charge at carbon, which is intercepted by the nucleophile. This provides a good explanation of what is going on here. The bromine begins to leave and a partial positive charge builds up at carbon. The departure of bromine can get to a more advanced state at the tertiary end than at the primary end because the substituents stabilize the build-up of positive charge. A more accurate representation of this bromonium ion is shown in the margin, with one C–Br bond longer than the other and more polarized than the other.

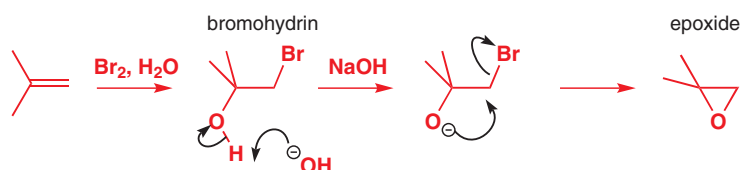


The nucleophile now has a choice: does it attack the more accessible, primary end of the bromonium ion, or does it attack the more charged end with the weaker C–Br bond? Here, the latter is clearly the faster reaction. The transition state has considerable positive charge on carbon and is known as a loose S_N2 transition state.



Interactive mechanism for regioselective addition to unsymmetrical alkenes

The products of bromination in water are called bromohydrins. They can be treated with base, which deprotonates the alcohol. A rapid intramolecular S_N2 reaction follows: bromide is expelled as a leaving group and an epoxide is formed. This can be a useful alternative synthesis of epoxides avoiding peroxy-acids.

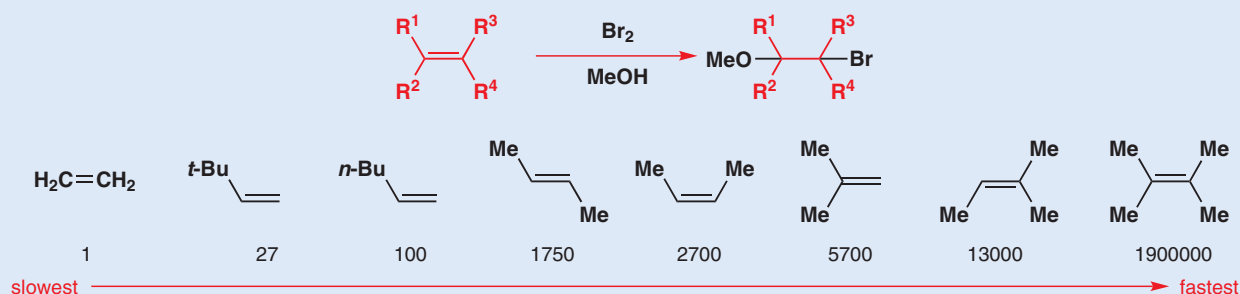


Rates of bromination of alkenes

The pattern you saw for epoxidation with peracids (more substituted alkenes react faster) is followed by bromination reactions too. The bromonium ion is a reactive intermediate, so the rate-determining step of the brominations is attack of bromine. The scale below shows the effect on the rate of reaction with bromine in methanol of increasing the number of alkyl substituents from none

(ethylene) to four. Each additional alkene substituent produces an enormous increase in rate. The degree of branching (Me versus *n*-Bu versus *t*-Bu) within the substituents has a much smaller, negative effect (probably of steric origin) as does the geometry (*E* versus *Z*) and substitution pattern (1,1-disubstituted versus 1,2-disubstituted) of the alkene.

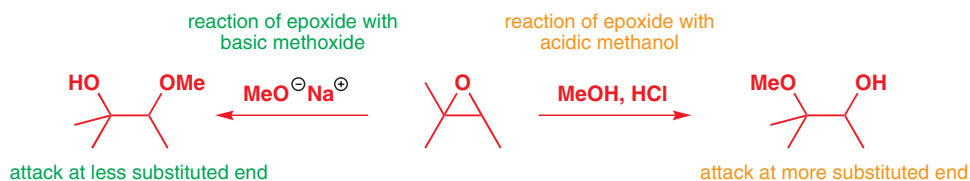
relative rates of reaction of alkenes with bromine in methanol



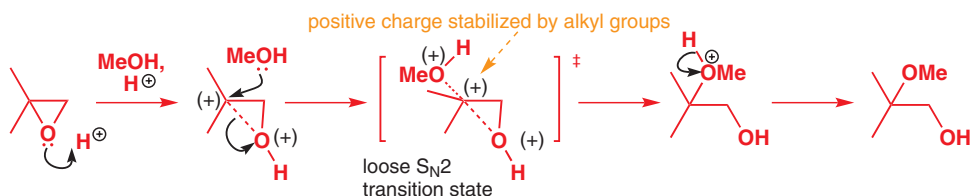
The regioselectivity of epoxide opening can depend on the conditions

Alkoxides are never leaving groups in S_N2 reactions: it's the strain alone which makes epoxides reactive.

Although epoxides, like bromonium ions, contain strained three-membered rings, they require either acid catalysis or a powerful nucleophile to react well. Compare these two reactions of a 1,1,2-trisubstituted epoxide. They are nucleophilic substitutions related to those we introduced in Chapter 15 (p. 352) but in that chapter we carefully avoided discussing epoxides of the unsymmetrical variety. In this example, the regiochemistry reverses with the reaction conditions. Why?

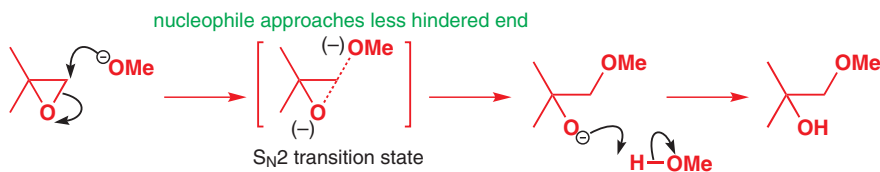


We'll start with the acid-catalysed reaction because it is more similar to the examples we have just been discussing—opening happens at the more substituted end. Protonation by acid produces a positively charged intermediate that bears a passing resemblance to the corresponding bromonium ion. The two alkyl groups make possible a build-up of charge on the carbon at the tertiary end of the protonated epoxide, and methanol attacks here, just as it does in the bromonium ion. You could think of the protonated leaving group 'pulling' the otherwise unreactive methanol in towards the reactive centre.



Remember, S_N1 can be fast only with good leaving groups (Chapter 15).

In base there can be no protonation of the epoxide and no build-up of positive charge. Without protonation, the epoxide oxygen is a poor leaving group, and leaves only if 'pushed' by a strong nucleophile: the reaction becomes pure S_N2 . Steric hindrance becomes the controlling factor and methoxide attacks only the primary end of the epoxide.



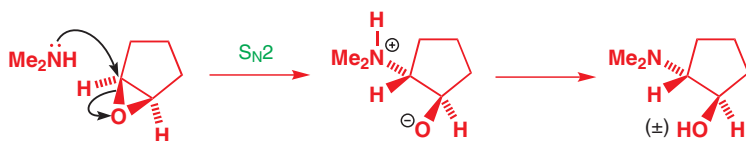
This example makes the matter look deceptively clear cut. But with epoxides, regioselectivity is not as simple as this because, even with acid catalysts, S_N2 substitution at a primary centre is fast. For example, Br^- in acid attacks the epoxide below mainly at the less substituted end, and only 24% of the product is produced by the 'cation-stabilized' pathway. It is very difficult to override the preference of epoxides unsubstituted at one end to react at that end.



For most substitution reactions of epoxides, then, regioselectivity is much higher if you give in to the epoxide's desire to open at the less substituted end and enhance it with a strong nucleophile under basic conditions.

Electrophilic additions to alkenes can be stereospecific

Although they really belong in Chapter 15 with other nucleophilic substitution reactions, we included the last few examples of epoxide-opening reactions here because they have many things in common with the reactions of bromonium ions. Now we are going to make the analogy work the other way by looking at the stereochemistry of the reactions of bromonium ions, and hence at the stereoselectivity of electrophilic additions to alkenes. We shall first remind you of an epoxide reaction from Chapter 15, where you saw this.

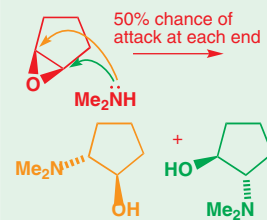


The epoxide ring opening is stereospecific: it is an S_N2 reaction and it goes with inversion. The epoxide starts on the top face of the ring and the amino group therefore ends up on the bottom face. In other words, the two groups end up *anti* or *trans* across the ring. You now know how to make this epoxide—you would use cyclopentene and *m*-CPBA, and in two steps you could ‘add’ an OH group and a Me_2N group *anti* across the double bond.



■ The reaction is stereospecific because it's the stereochemistry of the epoxide that determines the outcome of the reaction. The S_N2 reaction has no choice but to go with inversion. We discussed the terms ‘stereospecific’ and ‘stereoselective’ on p. 396.

■ Notice the $(\pm)$ symbol below the products. They are single diastereoisomers, but they are necessarily formed as racemic mixtures, as we discussed in Chapter 14. You can look at it this way: the Me_2NH will attack the two identical ends of the epoxide with precisely equal probability. Both give the same anti diastereoisomer, but each gives an opposite enantiomer. The two enantiomers will be formed in precisely equal amounts.



■ We don't need to write $(\pm)$ next to the isomer that isn't formed because it is an achiral structure—it has a plane of symmetry and would be a *meso* compound (see p. 317).

 Interactive mechanism for reaction of cyclohexene with bromine

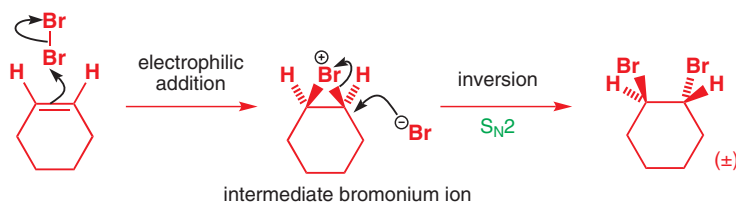
Now we can move on to look at the stereochemistry of electrophilic addition to alkenes.

Electrophilic addition to alkenes can produce stereoisomers

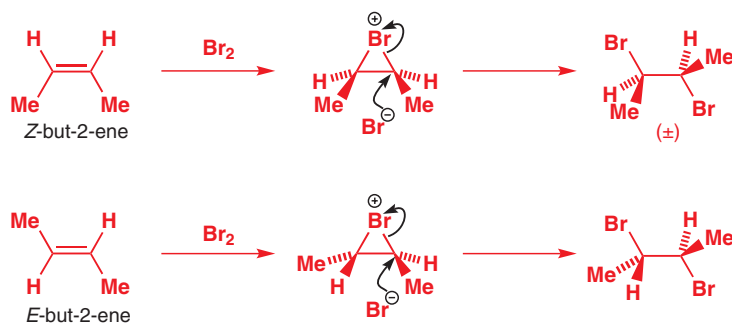
When cyclohexene is treated with bromine in carbon tetrachloride, the racemic *anti*-1,2-dibromocyclohexane is obtained exclusively.



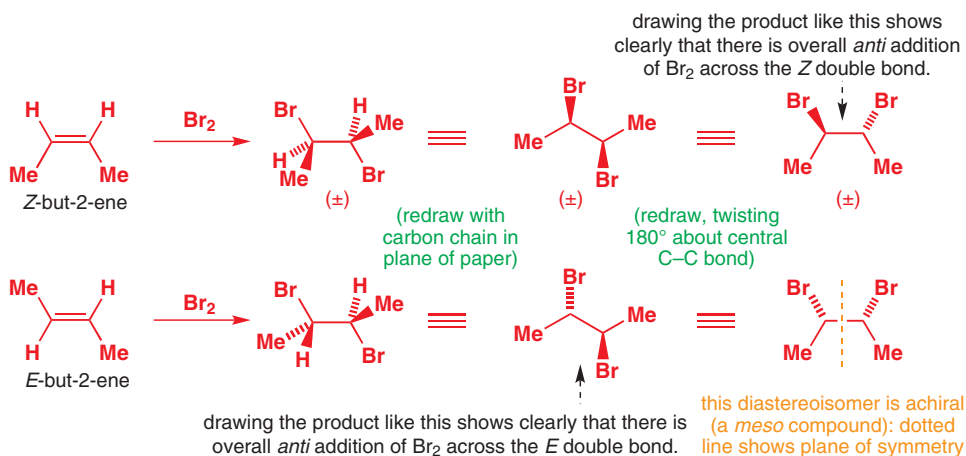
The result is no surprise if we think first of the formation of the bromonium ion that is opened with inversion in an S_N2 reaction.



Bromination of alkenes is stereospecific because the geometry of the starting alkene determines which product diastereoisomer is obtained. We couldn't demonstrate this with cyclohexene because only a *Z* double bond is possible in a six-membered ring. But bromination or chlorination of *Z* and *E*-2-butene in acetic acid produces a single diastereoisomer in each case, and they are different from each other. *Anti* addition occurs in both cases—more evidence that a bromonium ion is the intermediate.



The stereochemistry of the products is a bit clearer if we redraw them, and in the scheme below the product of each reaction is shown in two different ways. Firstly, the products have been rotated to place the carbon chain in the plane of the paper: in this conformation you can clearly see that there has been an *anti* addition across the *E* double bond. Secondly, the middle bond has been twisted 180° to give an (unrealistically) eclipsed conformation. We show this conformation for two reasons: now you can clearly see that there has been an *anti* addition across the *Z* double bond too. It also makes it quite clear that the product of the *E*-butene bromination is achiral: you can see the plane of symmetry in this conformation, and this is why we haven't placed (±) signs next to the products from the *E* alkene.



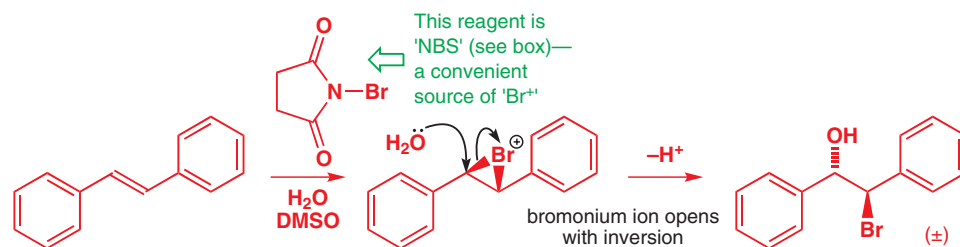
Interactive mechanism for stereospecific *anti* addition to alkenes

Note that in all three different views of each product the same stereoisomer is represented. There is no change of configuration, only changes of conformation to help you understand what is going on. If you cannot follow any of the 'redrawing' steps, make a model. With practice, you will soon learn to manipulate mental models in your head, and to see what happens to substituents when bonds are rotated. Most importantly, don't let all of this more subtle stereochemical discussion cloud the simple message:

● Bromine undergoes *anti* addition to alkenes.

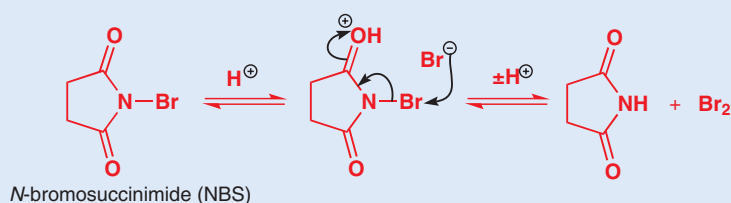
Bromonium ions as intermediates in stereoselective synthesis

You will not be surprised to learn that the other nucleophiles (water and alcohols) you saw intercepting bromonium ions earlier in the chapter also do so stereospecifically. The following reaction can be done on a large scale and produces a single diastereoisomer of the product (racemic, of course) because water opens the bromonium ion with inversion.

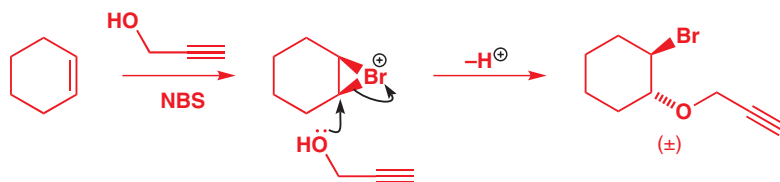


N-Bromosuccinimide, NBS

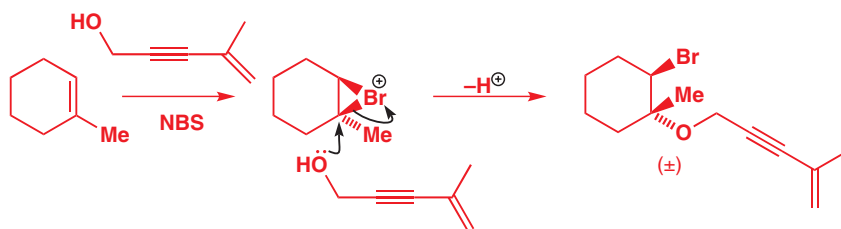
The reagent used to form the bromonium ion here is called *N*-bromosuccinimide, or NBS for short. Unlike the noxious brown liquid bromine, NBS is an easily handled crystalline solid and is perfect for electrophilic addition of bromine to alkenes when the bromonium ion is not intended to be opened by Br^- . It works by providing a very small concentration of Br_2 in solution: a small amount of HBr is enough to get the reaction going and thereafter every addition reaction produces another molecule of HBr , which liberates more Br_2 from NBS. In a sense, NBS is a source of Br^+ . NBS is known to act as a source of Br_2 because the results of reactions of NBS and of Br_2 in low concentration are identical.



The reagent NBS generates only a low concentration of Br_2 , so the concentration of Br^- is always low and alcohols compete with Br^- to open the epoxide even if they are not the solvent. In the next example, the alcohol is 'propargyl alcohol', prop-2-yn-1-ol. It gives the expected *anti*-disubstituted product with cyclohexene and NBS.

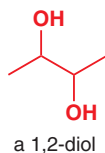


When 1-methylcyclohexene is used as the starting material, there is additionally a question of regioselectivity. The alcohol attacks the more hindered end of the bromonium ion—the end where there can be greatest stabilization of the partial positive charge in the 'loose $\text{S}_{\text{N}}2'$ ' transition state (see p. 437). This reaction really does illustrate the way in which a mechanism can lie in between $\text{S}_{\text{N}}1$ and $\text{S}_{\text{N}}2$. Configurational inversion, indicative of an $\text{S}_{\text{N}}2$ reaction, happens at a tertiary centre, where you would usually expect $\text{S}_{\text{N}}1$.



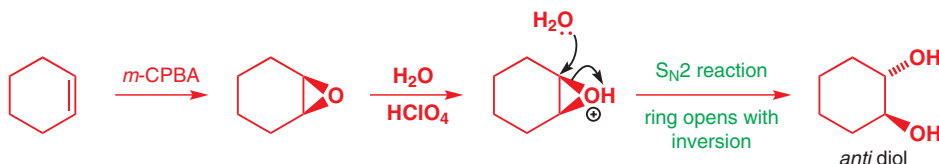
■ Notice that the bromine reacts only with the most electron-rich trisubstituted alkene and not with the disubstituted alkene or the alkyne.

Adding two hydroxyl groups: dihydroxylation



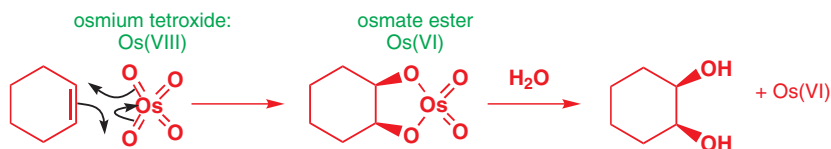
Many important compounds—the carbohydrates, for example—have two hydroxyl groups on adjacent carbon atoms. They are called 1,2-diols. A good way of making a 1,2-diol is to add two hydroxyl groups across a double bond. This can be done in two ways, each of which can give a different diastereoisomer of the product.

The first way uses chemistry you have already met. When a nucleophile opens an epoxide, it generates an alcohol. If the nucleophile is water, the product is the diol. The epoxide opening in an S_N2 reaction goes with stereochemical inversion, so in this example the two hydroxyl groups end up on opposite sides of the six-membered ring: the product is an *anti* diol. The epoxide opening reaction can be done in acid or in base.

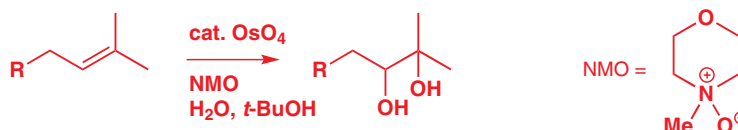


Interactive mechanism for dihydroxylation of alkenes

■ The mechanism of this reaction, in which the arrows go round in a ring and end where they started, is termed *pericyclic*: we shall discuss this in detail in Chapter 34.

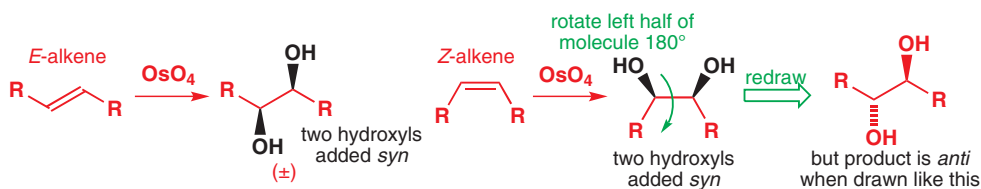


Because Os(VI) is produced in the reaction, and a simple oxidation will restore it to Os(VIII), the most effective version of this reaction makes use of just a catalytic amount of Os(VIII) and a stoichiometric amount of a reoxidant, often the compound NMO, or *N*-methylmorpholine-*N*-oxide. In the example below there is only one new chiral centre, so no possibility of diastereoisomers.



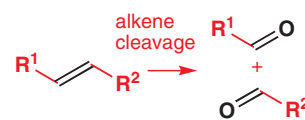
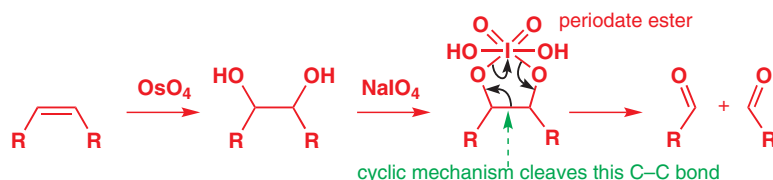
■ There is more discussion of this idea in the context of bromination on p. 440. It is worthwhile thinking about the chirality of these two products too: the first is chiral, with no plane of symmetry (we have included the symbol $(\pm)$ to remind you that although we have necessarily drawn just one enantiomer here, it must be racemic); the second is achiral, with a plane of symmetry in the first conformation shown and a centre of symmetry in the second. If this is not clear to you, look back at Chapter 15.

Because OsO_4 adds two hydroxyl groups to an alkene in a *syn* fashion, the overall product depends on the geometry of the alkene starting material: it is stereospecific. It is similar to bromination (p. 439) in that respect, although of course bromination is an *anti* addition. You can see how two different diastereoisomers are produced from different alkenes in these two examples: both dihydroxylations are mechanistically *syn*, but redrawing the product from the *Z* alkene in its more extended form reveals *anti* stereochemistry.



Breaking a double bond completely: periodate cleavage and ozonolysis

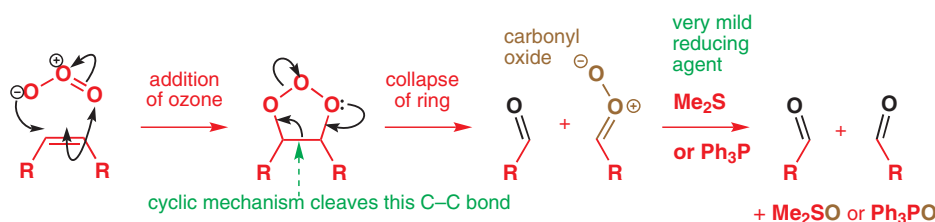
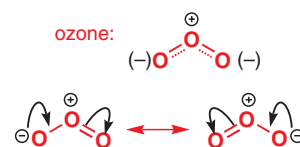
Sometimes it can be necessary to cleave a double bond completely, in other words to oxidize not just its π bond (as you have seen with Br_2 and OsO_4) but its σ bond too, as shown in the margin. This can be done in two steps using OsO_4 in conjunction with the reagent sodium periodate, NaIO_4 . The diol product forms a periodate ester, which decomposes to give two molecules of aldehyde by a cyclic mechanism similar to that for the OsO_4 step. The NaIO_4 also reoxidizes the Os(VI) to Os(VIII) so only a catalytic amount of Os is required.



Interactive mechanism for periodate cleavage of alkenes

The process proceeds by two successive oxidations—first of the π , and then the σ bond—with different reagents (which can be added in one step or in two—you can use NaIO_4 to cleave any diol, whether or not you made it using OsO_4). But there is another reagent that will achieve double oxidation in one step: ozone.

Ozone is a symmetrical bent molecule with a central positively charged oxygen atom and two terminal oxygen atoms that share a negative charge. Ozone is unstable, and is generated immediately before use from oxygen (using a device called an ‘ozonizer’) and bubbled into the reaction mixture. Like OsO_4 , it adds to alkenes by a cyclic mechanism: the product is a five-membered ring with three oxygen atoms. It is extremely unstable and collapses by breaking a weak O–O bond and a C–C σ bond, but gains two strong C=O bonds in the process.

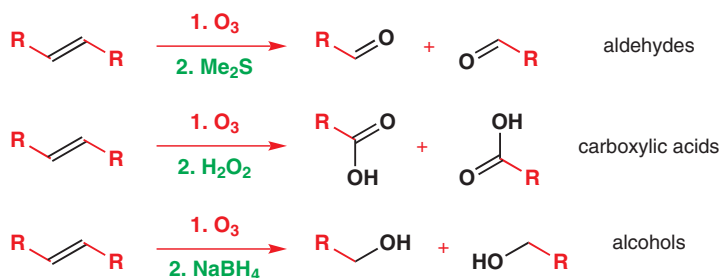


Interactive mechanism of ozonolysis

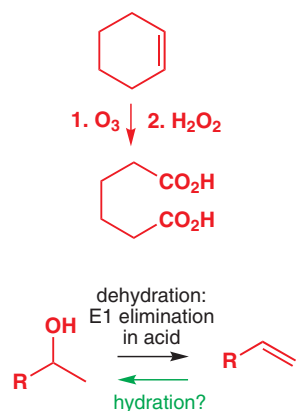
The immediate products are a simple aldehyde on the left and a new, rather unstable looking molecule known as a *carbonyl oxide* on the right. But treatment of this mixture with a very mild reducing agent such as dimethyl sulfide, Me_2S , or triphenylphosphine, Ph_3P , removes the ‘spare’ oxygen and reveals the two aldehydes.

This cleavage of an alkene by ozone is an important reaction and is known as *ozonolysis*. Ozonolysis can be used to generate not only aldehydes, but also other functional groups. Completing the reaction with oxidizing agents such as H_2O_2 will give carboxylic acids, and more powerful reducing agents such as NaBH_4 will give alcohols. Here are the overall transformations:

ozonolysis of alkenes to...



The mechanism by which the carbonyl oxide comes to be reduced is more complicated than we show here, and is addressed in Chapter 34.

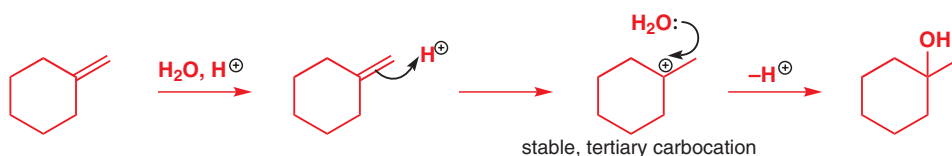


Ozonolysis of cyclohexenes is particularly useful as it gives 1,6-dicarbonyl compounds that are otherwise difficult to make. In the simplest case we get hexane-1,6-dioic acid (adipic acid), a monomer for nylon manufacture.

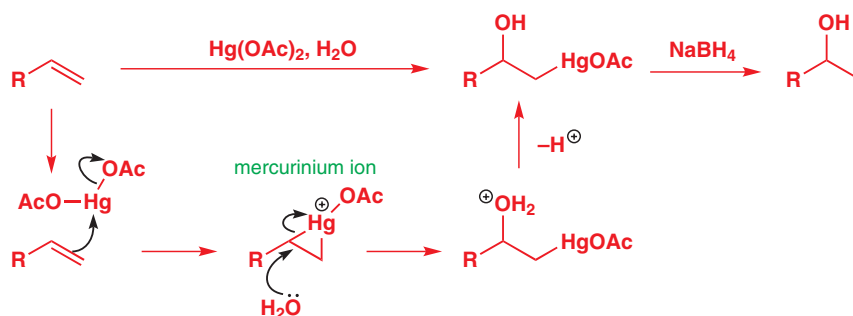
Adding one hydroxyl group: how to add water across a double bond

In Chapter 17 you saw alkenes being made from alcohols by E1 elimination—dehydration—under acid catalysis. The question we are going to answer in this section is: how can you make this elimination run backwards—in other words, how can you hydrate a double bond?

It is possible on occasion simply to use aqueous acid to do this. The reaction works only if protonation of the alkene can give a stable, tertiary cation. The cation is then trapped by the aqueous solvent.



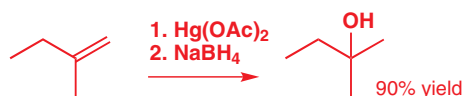
In general, though, it is difficult to predict whether aqueous acid will hydrate the alkene or dehydrate the alcohol. The method we are about to show you is much more reliable. The key is to use a transition metal to help you out. Alkenes are soft nucleophiles (p. 357) and interact well with soft electrophiles such as transition metal cations. In the margin, for example, is the complex formed between an alkene and mercury(II) cation. The complex should remind you of a bromonium ion, and rightly so because its reactions are similar. Even relatively feeble nucleophiles such as water and alcohols, when used as the solvent, open the ‘mercurinium’ ion and give alcohols and ethers. In the next scheme, the mercury(II) is supplied as mercury(II) acetate, $\text{Hg}(\text{OAc})_2$, which we shall represent with two covalent $\text{Hg}-\text{O}$ bonds. Unsurprisingly, water attacks at the more substituted end of the positively charged mercurinium ion.



■ The demercuration step involves radical chemistry, which is discussed in Chapter 37. You will find much more on organometallic compounds and their reactions in Chapter 40.

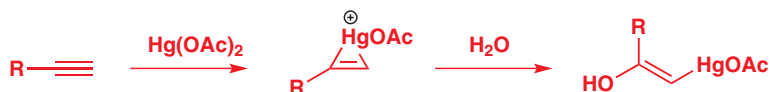
We’ve added OH and $\text{Hg}(\text{II})$ across the alkene, and the reaction is termed an ‘oxymercuration’. But a problem remains: how to get rid of the metal. The $\text{C}-\text{Hg}$ bond is very weak and the simplest way to replace Hg with H is by using a reducing agent: NaBH_4 works fine.

Below is an example of oxymercuration–demercuration at work. The intermediate mercury compound is not isolated.

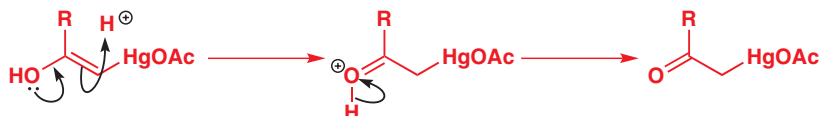


Hydration of alkynes

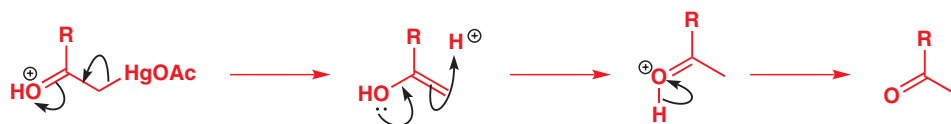
Oxymercuration works particularly well with alkynes. Here are the conditions, and the product, following the analogy of alkene hydration, should be the compound shown at the right-hand end of the scheme below.



But the product isolated from an alkyne oxymercuration is in fact a ketone. You can see why if you just allow a proton on this initial product to shift from oxygen to carbon—first protonate at C then deprotonate at O. C=O bonds are stronger than C=C bonds, and this simple reaction is very fast.



We now have a ketone, but we also still have the mercury. That is no problem when there is a carbonyl group adjacent because any weak nucleophile can remove mercury in the presence of acid, as shown below. Finally, another proton transfer (from O to C again) gives the real product of the reaction: a ketone.

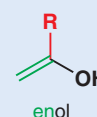


This is a very useful way of making methyl ketones because terminal alkynes can be made using the methods of Chapter 9 (addition of metallated alkynes to electrophiles).



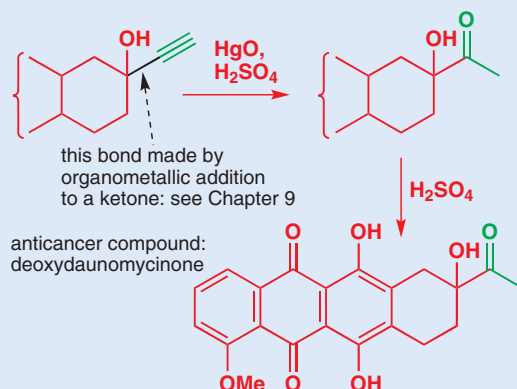
Enols

These alkenes carrying hydroxyl groups are called enols (ene + ol), and they are among the most important intermediates in chemistry. They happen to be involved in this reaction, and this was a good way to introduce you to them but, as you will see in the next chapter and beyond, enols (and their deprotonated sisters, enolates) have far-reaching significance in chemistry.

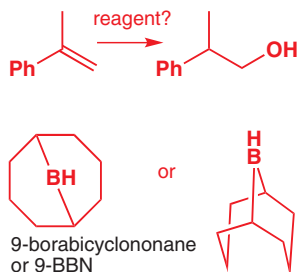


Anticancer compounds

The anthracycline class of anticancer compounds (which includes daunomycin and adriamycin) can be made using a mercury(II)-promoted alkyne hydration. You saw the synthesis of alkynes in this class on Chapter 9, where we discussed additions of metallated alkynes to ketones. Here is the final step in a synthesis of the anticancer compound deoxydaunomycinone: the alkyne is hydrated using Hg^{2+} in dilute sulfuric acid to give the final product.

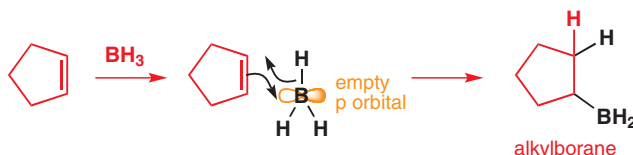


Hydroboration



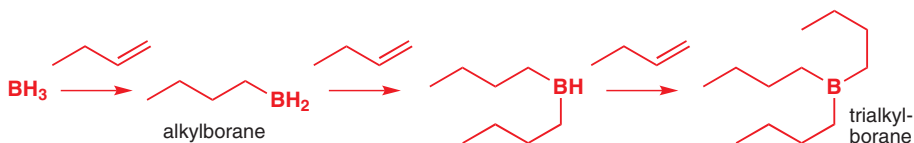
These methods for adding water across a double or triple bond involve cationic intermediates, and always end up putting the new hydroxyl group at the position best able to stabilize a positive charge (see p. 433). By what about addition of water the other way round? How would you do the reaction in the margin for example?

The answer is to make use of yet another element: boron. Boranes, including both BH_3 itself and analogues with one or two alkyl groups, HBR_2 (an important example is shown in the margin), add to alkenes to make a new C–H bond and a new C–B bond by a mechanism we can write like this. The alkene pushes electrons into the boron's empty p orbital, while the hydrogen transfers onto the alkene.

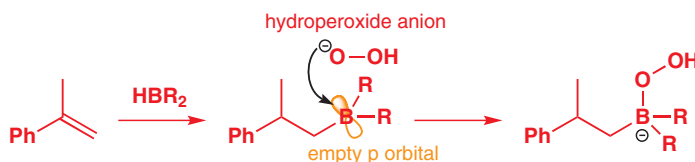


■ Why does B end up on the less substituted carbon? This is partly due to electronics—the reaction is driven by donation of the alkene's π electrons into the empty p orbital of the boron atom, so positive charge builds up at the other (more substituted) end of the alkene—and partly due to steric factors— BR_2 is bigger than H, so it ends up where there is less steric hindrance. One of the reasons for using the borane 9-BBN above is that the B atom is made very bulky by the bicyclic ring system.

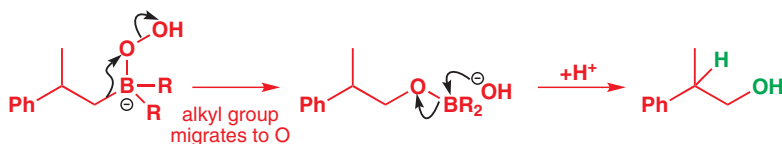
Importantly, if the alkene is unsymmetrical, the boron tends to end up on the less substituted carbon atom. This reaction can happen several times so, for example, if you start with an alkene and BH_3 , you will typically end up with a trialkylborane:



So far so good, if you want to make boranes, but we started out this section posing ourselves the problem of adding *water* across a double bond. This is where a quirk of boron chemistry helps us out. The C–B bond(s) we have just created can be oxidized to C–O bonds by using a mixture of NaOH and H_2O_2 . The mixture generates the hydroperoxide anion $\text{HO}-\text{O}^-$, which adds to that important empty p orbital on boron. The product is a negatively charged structure, shown below.



This is not stable, and it can decompose by a mechanism you should look at closely. It is not one which is familiar to you, but it makes sense if you think about it. The O–O bond is weak and can break, losing HO^- . As it does so, one of the alkyl groups on boron can migrate from B to O, relieving the boron atom of its negative charge, to give the structure shown below.



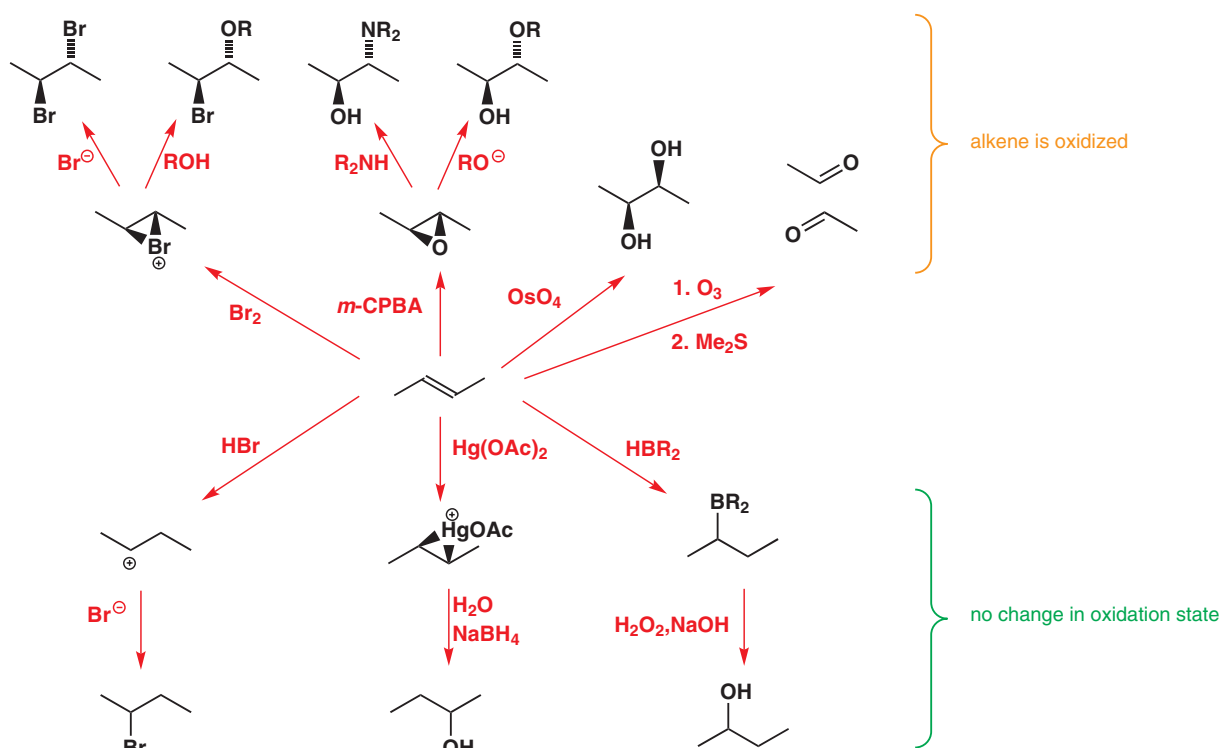
Interactive mechanism of hydroboration

➡ This mechanism, in which a C–B bond is exchanged for another C–X bond, is typical reactivity for boron. It has some similarity also with the Baeyer–Villiger oxidation, Chapter 36.

We now have the C–O bond where we want it, and all that has to happen is for the hydroxide anion to come back in and displace B from the alcohol product. The product, on protonation, is our alcohol. How can we be sure the correct R group will migrate? Well, if we use BH_3 we will get a trialkylborane, where all three groups on boron are the same, and all three C–B bonds can be oxidized one after another. If we use the HBR_2 reagent 9-BBN, then only the non-cyclic substituent formed in the hydroboration reaction will migrate, selectively giving us the product we want.

To conclude...a synopsis of electrophilic addition reactions

Electrophilic addition to double bonds gives three-membered ring intermediates with Br_2 , with Hg^{2+} , and with peroxy-acids (in which case the three-membered rings are stable and are called epoxides). All three classes of three-membered rings react with nucleophiles to give 1,2-difunctionalized products with control over (1) regioselectivity and (2) stereoselectivity. Protonation of a double bond gives a cation, which also traps nucleophiles, and this reaction can be used to make alkyl halides. Some of the sorts of compounds you can make by the methods of this chapter are shown below.



Further reading

The orbital arguments in this chapter are also treated in *Molecular Orbitals and Organic Chemical Reactions: Student Edition* by Ian Fleming, Wiley, Chichester, 2009. F. A. Carey and R. J. Sundberg, *Advanced Organic Chemistry A, Structure and Mechanisms*, 5th edn,

Springer, 2007, chapter 5, treats both elimination and addition reactions.

The stable bromonium ion on p. 428 was characterized by R. S. Brown et al., *J. Am. Chem. Soc.*, 1994, **116**, 2448.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>